

775.

TABLES OF COVARIANTS OF THE BINARY SEXTIC.

[Written in 1894: *now first published.*]

The binary sextic has in all (including the sextic itself and the invariants) 26 covariants which I have represented by the capital letters $A, B, C, \dots, Z$. The leading coefficients of the covariants A to R (of course for an invariant this means the invariant itself) are given in my paper “Tables for the binary sextic,” *Amer. Math. Jour.* vol. IV. (1881), pp. 379–384, [774]; the two invariants W and Z (Salmon’s invariants D and E) had been already calculated. But I did not in my values of the leading coefficients, nor did Salmon in his values of the two invariants, insert the literal terms with zero coefficients: as remarked in my paper [143] “Tables of the covariants M to W of the binary quintic,” it is very desirable to have in every case the complete series of literal terms, and I have accordingly in the expressions of the covariants A to R obtained for the leading coefficients, and in the expressions obtained from Salmon for the invariants W and Z , inserted in each case the complete series of literal terms.

I give a list of the 26 covariants nearly in the form of that given in the latter paper [143] for the covariants of the quintic, only instead of a separate column of deg-weights I insert these in the body of the symbol; thus

$$C = (3, 3, 4, 3, 3)^2 \text{ 4 to 8 } (x, y)^4,$$

the 5 coefficients of the quartic function contain respectively 3, 3, 4, 3, 3 terms (some of them it may be with zero coefficients), are of the degree 2, and of the weights 4, 5, 6, 7, 8 respectively.

The list is as follows:

$$\begin{aligned} A &= (1, 1, 1, 1, 1, 1)^1 \text{ 0 to 6 } (x, y)^6, \\ B &= (5)^2 \text{ 6 } (x, y)^0, \text{ Invt.}, \\ C &= (3, 3, 4, 3, 3)^2 \text{ 4 to 8 } (x, y)^4, \\ D &= (2, 2, 3, 3, 4, 3, 3, 2, 2)^2 \text{ 2 to 10 } (x, y)^8, \\ E &= (3, 3, 3, 3, 3)^3 \text{ 6 to 10 } (x, y)^4, \\ F &= (7, 7, 8, 8, 8, 7, 7)^3 \text{ 6 to 12 } (x, y)^6, \\ G &= (5, 7, 7, 8, 8, 8, 7, 7, 5)^3 \text{ 5 to 13 } (x, y)^8, \\ H &= (3, 4, 5, 7, 7, 8, 8, 8, 7, 7, 5, 4, 3)^3 \text{ 3 to 15 } (x, y)^{12}, \\ I &= (18)^4 \text{ 12 } (x, y)^0, \text{ Invt.}, \\ J &= (16, 16, 18, 16, 16)^4 \text{ 10 to 14 } (x, y)^4, \\ K &= (14, 16, 16, 18, 16, 16, 14)^4 \text{ 9 to 15 } (x, y)^6, \\ L &= (10, 13, 14, 16, 16, 18, 16, 16, 14, 13, 10)^4 \text{ 7 to 17 } (x, y)^{10}, \\ M &= (32, 32, 32)^5 \text{ 14 to 16 } (x, y)^2, \\ N &= (30, 32, 32, 32, 30)^5 \text{ 13 to 17 } (x, y)^4, \\ O &= (25, 29, 30, 32, 32, 32, 30, 29, 25)^5 \text{ 11 to 19 } (x, y)^8, \\ P &= (58)^6 \text{ 18 } (x, y)^0, \text{ Invt.}, \\ Q &= (51, 55, 55, 58, 55, 55, 51)^6 \text{ 15 to 21 } (x, y)^6, \\ R &= (51, 55, 55, 58, 55, 55, 51)^6 \text{ 15 to 21 } (x, y)^6, \\ S &= (94, 94, 94)^7 \text{ 20 to 22 } (x, y)^2, \\ T &= (90, 94, 94, 94, 90)^7 \text{ 19 to 23 } (x, y)^4, \\ U &= (151, 151, 147)^8 \text{ 23 to 25 } (x, y)^2, \\ V &= (221, 227, 227, 227, 221)^9 \text{ 25 to 29 } (x, y)^4, \\ W &= (338)^{10} \text{ 30 } (x, y)^0, \text{ Invt.}, \end{aligned}$$

$$X = (332, 338, 332)^{10} \text{ 29 to 31 } (x, y)^2,$$

$$Y = (668, 676, 668)^{12} \text{ 35 to 37 } (x, y)^2,$$

$$Z = (1636)^{15} \text{ 45 } (x, y)^0, \text{ Invt.}$$

$$A = (\mid x, y)^6$$

x^6	x^5y	x^4y^2	x^3y^3	x^2y^4	xy^5	y^6
$a + 1$	$b + 6$	$c + 15$	$d + 20$	$e + 15$	$f + 6$	$g + 1$

$$B = (\mid x, y)^0, \text{ Invt.}$$

$ag + 1$
$bf - 6$
$ce + 15$
$d^2 - 10$
± 16

$$C = (\mid x, y)^4$$

x^4	x^3y	x^2y^2	xy^3	y^4
$ae + 1$	$af + 2$	$ag + 1$	$bg + 1$	$cg + 1$
$bd - 4$	$be - 6$	$bf - 1$	$cf - 6$	$df - 4$
$c^2 + 3$	$cd + 4$	$ce - 9$	$de + 4$	$e^2 + 3$
		$d^2 + 8$		
± 4	± 6	± 9	± 6	± 4

$$D = (\mid x, y)^8$$

x^8	x^7y	x^6y^2	x^5y^3	x^4y^4	x^3y^5	x^2y^6	xy^7	y^8
$ac + 1$	$ad + 4$	$ae + 6$	$af + 4$	$ag + 1$	$bg + 4$	$cg + 6$	$dg + 4$	$eg + 1$
$b^2 - 1$	$bc - 4$	$bd + 4$	$be + 16$	$bf + 4$	$cf + 16$	$df + 4$	$ef - 4$	$f^2 - 1$
		$c^2 - 10$	$cd - 20$	$ce + 5$	$de - 20$	$e^2 - 10$		
			$d^2 - 20$					
± 1	± 4	± 10	± 20	± 20	± 20	± 10	± 4	± 1

$$E = (\mid x, y)^4$$

x^4	x^3y	x^2y^2
$ae g + 1$	$ad g + 1$	$ae^2 + 1$
$df - 3$	$ef - 1$	$f^2 - 1$
$e^2 + 2$	$a^2b^2cg - 1$	$a^2b^2dg - 3$
$a^2b^2d^2 - 1$	$bdf - 8$	$bef + 3$
$bcf + 3$	$be^2 + 9$	$c^2g + 2$
$bde - 1$	$cdf + 9$	$cdf - 1$
$c^2e - 3$	$cde - 17$	$ce^2 - 3$
$cd^2 + 2$	$d^3 + 8$	$d^2e + 2$
± 3	± 1	± 1
± 5	± 26	± 7

$$F = (\mid x, y)^6$$

x^6	x^5y	x^4y^2	x^3y^3	x^2y^4	xy^5	y^6
$a^2cg \dots$	$abg \dots$	$ae g + 1$	$ad g + 2$	$ae^2 + 1$	$af g \dots$	$ag^2 \dots$
$abf \dots$	$cf + 2$	$df + 2$	$ef - 2$	$f^2 - 1$	$a^2beg + 2$	$a^2bfg \dots$
$ce + 1$	$-2de$	$e^2 - 3$	$a^2c^2g - 2$	$bf^2 - 2$	$ceg + 1$	
$d^2 - 1$	$a^2b^2g - 1$	$a^2b^2cg - 1$	$bdf + 4$	$bef - 2$	$cdg - 2$	$cf^2 - 1$
$-a^2b^3e - 1$	$a^2bf - 2$	$bcf - 2$	$be^2 - 2$	$c^2f - 3$	$cef + 2$	$d^2g - 1$
$bcd + 2$	$bce + 2$	$bde + 4$	$c^2f - 2$	$cdf + 4$	$def + 2$	$def + 2$
$c^3 - 1$	$a^2c^2d - 2$	$c^2d + 2$	$cde + 6$	$ce^2 + 2$	$de^2 - 2$	$e^3 - 1$
		$-3d^3$	$-d^3 - 2$	$d^2e - 3$		
± 1	± 2	± 3	± 2	± 1	± 6	± 3
± 2	± 4	± 6	± 10	± 8		

The table for $G = (\mid x, y)^8$ is rotated 90° in the scan; it consists of nine columns indexed by $x^{8-k}y^k$ for $k = 0, \dots, 8$, each column listing the literal monomials in a, b, c, d, e, f, g of degree 3 and the appropriate weight, with the column sums tabulated at the foot. The complete literal series (with zero coefficients restored) is given on this page of the scan; for reasons of width the present transcription reproduces only the leading column entries and column totals as verified from the PNG:

Column x^8 leading terms: $adf + \dots, ae^2 \dots, bcf \dots, bde \dots, c^2e \dots, cd^2 \dots$; column total ± 11 .

Column x^7y leading terms: $aeg \dots, af^2 \dots, bdg \dots, bef \dots, c^2g \dots, cdf \dots, ce^2 \dots, d^2e \dots$; column total ± 60 .

Column x^6y^2 : leading $afg \dots, beg \dots, bf^2 \dots, cdg \dots, cef \dots, d^2f \dots, de^2 \dots$; column total ± 110 .

Column x^5y^3 : leading $ag^2 \dots, bfg \dots, ceg \dots, cf^2 \dots, d^2g \dots, def \dots, e^3 \dots$; column total ± 180 .

Column x^4y^4 (centre): $bg^2 \dots, cfg \dots, deg \dots, df^2 \dots, e^2f \dots$; central column total ± 270 .

Symmetric columns $x^3y^5, x^2y^6, xy^7, y^8$ mirror the weights of columns $x^5y^3, x^6y^2, x^7y, x^8$ respectively under $a \leftrightarrow g, b \leftrightarrow f, c \leftrightarrow e, d \rightarrow d$, with totals $\pm 180, \pm 110, \pm 60, \pm 11$.

[Tables for G on PDF p. 404 (printed 380); the full coefficient detail in the scan occupies the entire page at rotated orientation and is best consulted against the facsimile.]

$$I = (\mid x, y)^0, \text{ Invt.} \quad J = (\mid x, y)^4$$

I	x^4	x^3y	x^2y^2	xy^3	y^4
$ace^2 \dots$	$adg \dots$	$a^2fg + 2$	$a^2g^2 + 1$	$abg^2 + 2$	$acg^2 \dots$
$af^2 \dots$	$ae f - 10$	$ae fg - 10$	$afg - 6$	$cfg - 10$	$dfg \dots$
$d^2f \dots$	$bcg - 1$	$bf^2 + 8$	$beg - 6$	$deg + 4$	$e^2g \dots$
$def + 2$	$bdf - 8$	$cdg + 4$	$cf^2 + 6$	$df^2 + 16$	$ef \dots$
$e^2 - 1$	$be^2 + 9$	$cef + 26$	$de^2 + 8$	$ef^2 - 12$	$a^2bg^2 + 1$
$a^2b^2g - 1$	$c^2f + 9$	$df + \dots$	$d^2g + 4$	$a^2bfg - 8$	$bcfg - 10$
$bf + 1$	$cde - 17$	$de^2 - 8$	$def + 12$	$bceg + 26$	$bdeg + 4$
$dg - 1$	$d^3 + 8$	$a^2b^3dg + 16$	$e^3 - 12$	$bcf^2 + 24$	$bcf^2 + 24$
$def + 2$	$a^2b^2eg + 16$	$bf + \dots$	$a^2b^2fg + 6$	$bd^2g - 8$	$bdf^2 + 16$
$e^2 - 1$	$b^2f + 9$	$bc^2g + 24$	$b^2f + \dots$	$bdef - 64$	$bef - 12$
$bcdg + 2$	$bcf - 12$	$bc^2g - 12$	$bcdg + 12$	$be^2 + 36$	$c^2eg + 16$
$bcef - 2$	$bcde - 76$	$bcdf - 64$	$bcef + 18$	$c^2dg - 8$	$c^2dg - 8$
$bdf - 2$	$bd^3 + 48$	$bce^2 - 42$	$bd^2f - 96$	$c^2ef - 42$	$cdf + 9$
$bde^2 + 2$	$c^3e + 48$	$bd^2e + 56$	$bde^2 + 60$	$cd^2e + 4$	$cd^2g - 12$
$c^3g - 1$	$c^2d^2 - 32$	$c^3f + 36$	$c^3g - 12$	$d^3e - 16$	$cdef - 76$
$c^2df + 2$		$c^2d^2 + 4$	$c^2df + 60$		$ce^3 + 48$
$c^2e^2 + 1$			$c^3 - 99$		$d^2f + 48$
$-3cd^2e$			$cd^2 + 84$		$c^3d^2 - 32$
$d^4 + 1$			$d^4 - 32$		
-3	± 142	± 168	± 263	± 168	± 142
$+9$					

$$K = (\mid x, y)^6$$

x^6	x^5y	x^4y^2	x^3y^3	x^2y^4	xy^5	y^6
$a^2dg + 1$	$a^2eg + 2$	$a^2fg \dots$	$a^2g^2 \dots$	$a^2g^2 \dots$	$aeg^2 - 2$	$adg^2 - 1$
$ae f - 1$	$f^2 - 2$	$a^2beg + 10$	$abfg \dots$	$cfg - 10$	$dfg + 2$	$cfg + 3$
$abcg - 3$	$a^2bdg - 2$	$bf^2 - 10$	$ceg \dots$	$deg + 15$	$e^2f + 6$	$e^2f - 2$
$bdf - 2$	$bef + 2$	$cdg - 15$	$cf - 20$	$df^2 + 10$	$ef - 6$	$a^2bcg^2 + 1$
$be^2 + 5$	$c^2g - 6$	$cef - 5$	$d^2g + \dots$	$ef^2 - 15$	$a^2bf^2 + 2$	$a^2bdg^2 + 2$
$bf + 9$	$cdf + 28$	$df + 60$	$def - 60$	$a^2bfg + 10$	$bcfg - 2$	$bcfg - 9$
$cde - 17$	$ce^2 - 26$	$d^3 - 40$	$e^3 - 40$	$bceg + 5$	$bdeg - 28$	$bdef + \dots$
$d^2 + 8$	$d^2e + 4$	$a^2b^2dg - 10$	$a^2b^2eg + 20$	$bcf^2 - 30$	$bdf^2 + 32$	
		$b^2ef + 30$	$bf + \dots$	$bd^2g - 60$	$bef - 6$	$b^2f^2 + 6$
$a^2b^2cg + 2$	$a^2b^2cg - 6$			$bdef + 110$	$c^2eg + 26$	
		$bc^2g + 15$	$bcdg - 60$	$bef - 45$	$c^2f - 36$	$cdg^2 - 5$
$b^2cf - 6$	$b^2df - 32$	$bcdf - 110$		$cdg + 40$	$cd^2g - 4$	
$b^2de + 2$	$b^2e^2 + 36$	$bcf^2 + 15$	$bcef + \dots$	$cef - 15$		$cdeg + 17$
$bc^2e + 6$	$bcf + 6$	$bd^2g + 40$	$bdf + \dots$	$cdf - 40$		$cdef + 58$
$bcd^2 - 4$	$bcde - 58$	$cf + 45$	$bde^2 + 20$	$cde^2 + 25$	$df - 32$	$cd^2 - 2$
$c^3d \dots$	$bd^3 + 32$	$c^2de - 25$	$c^3g + 40$		$d^2e^2 + 20$	$cef - 6$
	$c^3e + 30$		$c^2df - 20$			$a^2g - 8$
	$c^2d^2 - 20$	$cd^3 \dots$	$c^2e^2 \dots$	$d^3e \dots$		$b^2f^2 + 4$
			$cd^2e \dots$			$b^2e^2 \dots$
			$d^4 \dots$			
$+33$	± 146	± 215	± 140	± 215	± 146	± 33

[Page 382 of the scan presents the table $L = (\mid x, y)^{10}$ rotated 90° to the page, an 11-column array of coefficients of degree 4 and weights 7 through 17. The columns are indexed by $x^{10-k}y^k$ for $k = 0, \dots, 10$; under the weight-substitution $a \leftrightarrow g, b \leftrightarrow f, c \leftrightarrow e, d \rightarrow d$ the array is symmetric about its centre column x^5y^5 . The complete listing is best consulted against the facsimile.]

Column totals (from foot of the rotated table, read left to right):

$$\pm 17, \pm 60, \pm 110, \pm 270, \pm 376, \pm 294, \pm 376, \pm 270, \pm 110, \pm 60, \pm 17.$$

Leading column (x^{10} , weight 7) entries:

$$a^2df^2 + 1, \quad a^2e^2f \dots, \quad a^2 \text{ terms} \dots, \quad bcf \dots, \quad bd^2 \dots, \quad c^2 \text{ terms} \dots$$

(complete listing in scan; see facsimile p. 382).

$$M = (\mid x, y)^2$$

x^2	xy	y^2
$a^2cg^2 + 1$	$a^2bg^2 \dots$	$a^2eg^2 + 1$
$dfg - 6$	$cfg + 8$	$a^2fg \dots$
$e^2g + 8$	$efg + 8$	$f^2g - 1$
$ef^2 - 3$	$f^3 - 6$	$a^2bdg^2 - 6$
$a^2bfg - 1$	$a^2bcg^2 + 8$	$a^2befg + 6$
$bcfg + 6$	$bdfg - 20$	$bf^2 \dots$
$bdeg - 34$	$be^2g - 24$	$c^2g + 8$
$bdf + 48$	$bef^2 + 36$	$cdfg - 34$
$be^2 - 18$	$c^2fg - 24$	$ce^2g + 18$
$c^2eg + 18$	$cdfg - 24$	$cef^2 \dots$
$c^2f^2 - 45$	$cdeg + 76$	$d^2eg + 4$
$cdf + 4$	$cd^2 + 36$	$d^2f + 64$
$cdef + 78$	$ce^3 - 72$	$de^2f - 96$
$ce^2 - 36$	$cef - 72$	$e^4 + 36$
$d^2f - 48$	$d^2g - 32$	$a^2b^2cf - 3$
$d^2e + 28$	$d^2f - 8$	$b^2cfg - 48$
$a^2b^2cf - 6$	$de^2 + 24$	$b^2cfg - 45$
$b^2cdg + 36$	$a^2b^2cf - 6$	$b^2f^2 \dots$
$b^2deg + 36$	$b^2cdg + 36$	$bdfg - 18$
$b^2f^2 \dots$	$b^2df^2 \dots$	$bcdeg + 78$
$b^2c^2g + 64$	$b^2e^2f - 54$	$bcdf - 144$
$b^2cef - 144$	$bc^2ef - 72$	$bce^3 + 108$
$b^2e^2 + 81$	$bc^2f - 54$	$bd^2g - 48$
$bc^2dg - 96$	$bcd^2g - 8$	$bd^2f + 96$
$bce^2f + 108$	$bcdef - 36$	$bde^2 - 72$
$bcdf^2 + 96$	$bce^3 + 216$	$c^2eg - 36$
$bcde^2 - 126$	$bd^3f + 128$	$c^2f + 81$
$bd^3e + 16$	$bd^2e^2 - 192$	$c^2d^2g + 28$
$c^4g + 36$	$c^2d^2g + 24$	$c^2def - 126$
$c^3df - 72$	$c^3ef + 216$	$c^3d^2f - 192$
$c^3e^2 - 27$	$c^3d^2f - 192$	$c^2d^3 - 27$
$c^2d^3 + 96$	$c^2de^2 - 378$	$cd^2f + 16$
$cd^4 - 32$	$cd^3e + 464$	$cd^2e^2 + 96$
	$d^5 - 128$	$d^4e - 32$
+9	−8	+1
182		±136
497	±1120	±551
±688	±1308	±688

$$N = (|x, y)^4$$

x^4	x^3y	x^2y^2	xy^3	y^4
$a^2bg^2 \dots$	$a^2cg^2 - 1$	$a^2dg^2 \dots$	$a^2eg^2 + 1$	$a^2fg^2 \dots$
$cfg - 1$	$dfg + 4$	$efg - 1$	$f^2g - 1$	$abeg^2 + 1$
$deg + 1$	$e^2g \dots$	$f^2 - 3$	$a^2bdg^2 - 4$	$b f^2g - 1$
$df^2 + 3$	$ef^2 - 3$	$a^2bcg^2 - 3$	$b^2fg + 4$	$c dg^2 - 1$
$ef - 3$	$a^2b^2g^2 + 1$	$bdfg \dots$	$bf^3 \dots$	$c efg - 2$
$e^2 \dots$	$b^2fg - 4$	$be^2g - 15$	$c^2f \dots$	$c f^2 + 3$
a^2b^2cg	$bdeg - 16$	$bef + 18$	$cdfg + 16$	$c^2fg + 4$
$bceg + 2$	$bdf^2 \dots$	$c^2fg + 15$	$c^2g - 22$	$de^2g - 1$
$bcf - 3$	$b^2f + 18$	$cdeg \dots$	$cef^2 + 6$	$d^2f^2 - 6$
$bd^2g - 4$	$c^2eg + 22$	$cdf^2 - 36$	$d^2eg + 8$	$ef + 3$
$bdef - 12$	$c^2f^2 + 3$	$cef + 9$	$df - 32$	$a^2b^2df - 3$
$be^3 + 15$	$cd^2g - 8$	$df \dots$	$d^2f + 36$	$b^2efg + 3$
$c^2dg + 1$	$cdef - 48$	$d^2f + 24$	$e^4 - 12$	$b f^3 \dots$
$c^2ef + 9$	$ce^2 + 12$	$d^2f - 12$	$a^2b^2cg - 3$	$b c^3f + 3$
$cdf^2 + 4$	$d^2f + 32$	$a^2b^2fg + 3$	$b^2c^2g \dots$	$b cdfg + 12$
$cd^2 - 21$	$d^2e^2 - 12$	$b^2ef - 18$	$b^2f - 3$	$b c^2g - 9$
$d^4 + 8$	$a^2b^2f^2 \dots$	$b^2deg + 36$	$b^2ef \dots$	$b cef - 9$
$a^2b^3eg - 3$	$a^2b^2cf \dots$	$b^2cf \dots$	$b^2cg - 18$	$b d^2eg - 4$
$b^2f^2 \dots$	$b^2cef - 6$	$b^2cef - 27$	$bcdeg + 48$	$b df^2 - 32$
$b^2cdg + 6$	$b^2d^2g + 32$	$b^2c^2eg - 9$	$bcd f^2 \dots$	$b d^2f + 66$
$b^2cef + 9$	$b^2def \dots$	$b^2cf^2 + 27$	$bce^2 - 18$	$b e^2 - 27$
$b^2df^2 + 32$	$b^2e^2 - 27$	$b^2cd^2g - 24$	$bd^2g - 32$	$c^2efg - 15$
$b^2de^2 - 39$	$b^2c^2dg - 36$	$b^2cdef \dots$	$b^2d^2f + 32$	$c^2deg + 21$
$b c^2g - 3$	$b^2c^2ef + 18$	$b^2cef + 27$	$bde^3 - 12$	$c^2df^2 + 39$
$b c^2df - 66$	$b^2cd^2f - 32$	$b^2cd^2f \dots$	$c^2eg - 12$	$c^2ef - 18$
$b c^2f + 18$	$b^2cde^2 + 84$	$b^2d^2e^2 - 12$	$df + 27$	$c d^2g - 8$
$b cd^2e + 76$	$b^2d^2e - 32$	$b c^2dg^2 + 12$	$c^2d^2g + 12$	$cd^2ef - 76$
$b d^4 - 32$	$c^4g + 12$	$b c^2ef - 27$	$c^2def - 84$	$cd^2e^2 + 45$
$c^4f + 27$	$c^4df + 12$	$b c^2df + 12$	$c^3f + 45$	$d^4f + 32$
$c^3de - 45$	$-45c^3e^2$		$cd^2f + 32$	$d^4e^2 - 20$
$c^2d^3 + 20$	$c^3d^2 + 20$	$c^2de^2 \dots$	$cd^3e^2 - 20$	
	$c^2d^3 \dots$	$c^2d^2e \dots$	$d^4e \dots$	
		$cd^3 \dots$	d^5	

[Page 385 presents the table $O = (\mid x, y)^8$, a 9-column array indexed by $x^{8-k}y^k$ ($k = 0, \dots, 8$) of coefficients of degree 5 and weights 11 through 19. The complete listing occupies the full page; column totals were not separately tabulated in the scan.]

Selected leading entries by column:

x^8 : $a^2fg^2 \dots, a^2beg^2 \dots, cfg^2 - 3, efg - 5, f^2 \dots, \dots$ (full series in scan).

x^7y : $a^2bfg^2 \dots, abf^2g \dots, cef^2 \dots, df^3 + 2, e^2f^2 - 10, \dots$

x^6y^2 : leading $ae fg^2 \dots, bf^2g - 3, cd fg + 10, defg - 5, ef^3 \dots$

x^5y^3 : leading $a^2g^3 \dots, a^2bfg^2 \dots, beg^2 - 5, cf^2g - 7, d^2fg + 2, de^2g - 10$

x^4y^4 (centre): $a^2f^2g \dots, abefg + 5, cef^2 - 12, deg^2 + 10, e^2fg - 5$

Subsequent columns $x^3y^5, x^2y^6, xy^7, y^8$ are obtained from the four leading columns by the duality $a \leftrightarrow g, b \leftrightarrow f, c \leftrightarrow e, d \leftrightarrow d$.

[Full coefficient detail in scan p. 385 (printed); see facsimile.]

$$P = (\mid x, y)^0, \text{ Invt.}$$

$a^2b^2g^4 \dots$
$a^2bfg^3 \dots$
$a^2b^2eg^3 \dots$
$cf^3 \dots$
$d^2g^2 + 1$
$defg - 6$
$df^2 + 4$
$e^2g + 4$
$ef^2 - 3$
$ab^2efg^2 \dots$
$f^3g \dots$
$abcdg^2 - 6$
$cefg + 18$
$cf^2 - 12$
$d^2fg + 12$
$d^2eg - 18$
$de^2f + 6$
$b^2cf + 4$
$a^2b^2dfg - 18$
$c^2e^2g - 24$
$c^2ef^2 + 30$
$cd^2eg + 54$
$cdf^2 - 12$
$cdeg^2 - 42$
$ce^3 + 12$
$d^3g - 20$
$d^2ef + 24$
$d^2 - 8$
$a^2b^2dg^2 + 4$
$efg - 12$
$f^2 + 8$
$b^2fg^2 - 3$
$cdfg \dots$
$ce^2g + 30$
$cef^2 - 24$
$d^2fg - 12$
$df^2 - 24$
$de^2f + 60$
$e^4 - 27$
$bc^3fg + 6$
$c^2deg - 42$
$c^2df^2 + 60$
$c^2e^2f - 30$
$cd^2g + 24$
$cdf^2 - 84$
$cde^2 + 66$
$d^4f + 24$
$d^4e^2 - 24$
$b^2c^2eg + 12$
$cf^2 - 27$
$-c^2d^2g \ 8$
$c^2def + 66$
$c^2e^2 - 8$
$cdf - 24$
$c^2d^2e^2 - 39$
$cd^2e + 36$
$d^4 - 8$

[Page 387 presents the table $Q = (\mid x, y)^6$, a 7-column array indexed by the column headers 51, 55, 55, 58, 55, 55, 51 at the top—these are the term-counts per column—and $x^{6-k}y^k$ for $k = 0, \dots, 6$. Each column lists the literal monomials of degree 6 and weights 15 through 21. The full content spans this page and the next (p. 388, marked “ Q (continued)”), and is best consulted against the facsimile.]

Selected leading entries from the principal columns of Q :

Column 1 (x^6 , header 51, weight 15): $a^2bdg^3 \dots, efg \dots, f^2 - 2, a^2bcf + 5, dfg \dots, e^2 - 8, a^2b^2cf - 2, cdfg + 6, c^2g + 8, cef - 10, -d^2eg + 10, df + 18, de^3 - 22, e^4 + 12, ab^2 + 4, dfg - 6, e^2g - 8, \dots$; column total ± 110 .

Column 2 (x^5y , header 55, weight 16): $a^2beg^3 \dots, f^2g \dots, bef \dots, cdfg + 6, c^2f + 8, cef - 10, d^2eg + 18, def - 22, e^3 + 12, abf^2 + 4, \dots$

Column 3 (x^4y^2 , header 55, weight 17): $a^2bfg^3 \dots, a^2b^2eg^2 \dots, cefg + 15, cf^2 - 10, dg - 15, dfg + 20, de^2g - 25, def - 10, e^2f + 15, a^2bf + 4, bdfg + 5, bef - 90, ce^2g + 40, cef^2 + 40, d^2fg + 50, df^2 + 10, de^2f - 40, e^4 - 15, bdef + 35, c^2deg - 45, \dots$

Column 4 (x^3y^3 , header 58, weight 18; centre): $a^2fg^3 \dots, a^2bfg^3 \dots, cf^2g \dots, df^2 \dots, defg + 20, e^2fg - 20, ef^3 + 20, a^2b^2fg^2 - 6, c^4g \dots, c^2f^2 - 20, c^2deg \dots, c^2df + 60, c^2e^3 - 20, cd^2eg \dots, cd^2f + 62, cd^2 - 60, bcd^2f + 66, bcde - 126, bd^3e + 24, c^4f - 50, c^3e + 60, c^2d^3 - 30, c^2de^2 + 60, c^2d^2 - 15, cd^3 \dots, d^6 \dots$

Columns 5, 6, 7 (x^2y^4 , xy^5 , y^6) are the weight-mirrors of columns 3, 2, 1 under the involution $a \leftrightarrow g$, $b \leftrightarrow f$, $c \leftrightarrow e$, $d \rightarrow d$.

[Full coefficient detail for table Q on facsimile pp. 387–388.]

$Q = (\mid x, y)^6$ (continued).

[Continuation of the table Q from p. 387 above; remaining literal monomials for each of the 7 columns and column totals. See facsimile.]

Selected continued entries:

Column 1 (continued, x^6): $a^2b^2c^2g + 76, cdef + 125, b^2 - 24, d^3 \dots, bcdef + 105, b^2ce^2 - 44, \dots, a^2b^2fg - 2, a^2b^2cg - 4, \dots, bdcef + 50, cdf - 44, -c^2d^2eg - 112, c^2d^3 - 12, c^2df + 28, cd^2e \dots, d^5 + 32, cd^4 \dots$

Column 4 (centre, x^3y^3 , continued): $bc^2d^2f + 74, a^2d^4 + 22, \dots, cd^3 + 10, c^4 - 25, d^3 \dots$

Column 7 (y^6 , continued, mirror of column 1): g -terms mirror a -terms above.

[End of paper 775; column totals at foot of p. 388 of scan.]

776.

ON THE JACOBIAN SEXTIC EQUATION.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. XVIII. (1882), pp. 52–65.]

The Jacobian sextic equation has been discussed under the form

$$(z - \alpha)^6 - 4a(z - \alpha)^5 + 10b(z - \alpha)^3 - 4c(z - \alpha) + 5b^2 - 4ac = 0,$$

(see references at end of paper), but the connexion of this form with the general sextic equation has not, so far as I am aware, been considered. And although this is probably known, I do not find it to have been explicitly stated that the group of the equation is the positive half-group, or group of the 60 positive substitutions out of the 120 substitutions, which leave unaltered Serret's 6-valued function of six letters.

Invariantive Property of the Jacobian Sextic.

Taking $z - \alpha$ as the variable, and comparing the equation with the general sextic equation

$$= (a, b, c, d, e, f, g \mid z - \alpha, 1)^6 0,$$

we have

$$a, b, c, d, e, f, g = 1, -\frac{2}{3}a, 0, \frac{1}{2}b, 0, -\frac{2}{3}c, 5b^2 - 4ac;$$

the Jacobian equation is thus an equation

$$(a, b, c, d, e, f, g \mid x, y)^6 = 0,$$

for which $c = 0, e = 0, ag + 9bf - 20d^2 = 0$; but of course any equation, which can be by a linear transformation upon the variables brought into this form, may be regarded as a Jacobian equation.

Hence, using henceforward the small italic in place of the small roman letters, the Jacobian sextic may be regarded as an equation

$$= (a, b, c, d, e, f, g \mid x, y)^6 0,$$

linearly transformable into the form

$$= (a, b, 0, d, 0, f, g \mid x, y)^6 0,$$

where $ag + 9bf - 20d^2 = 0$. It is to be shown, that this implies a single relation between the four invariants A, B, C , and D of the sextic function.

I call to mind that the general sextic has five invariants A, B, C, D, E of the orders 2, 4, 6, 10, 15 respectively; the last of them E is not independent, but its square is equal to a rational and integral function of A, B, C, D ; and instead of D , we consider the discriminant Δ which is an invariant of the same order 10. The values of A, B, C are given. Table Nos. 31, 34, and 35 of my Third Memoir on Quantics, *Phil. Trans.*, vol. CXLVI. (1856), pp. 627–647, [144]; those of D, Δ, E were obtained by Dr Salmon, see his *Higher Algebra*, second ed. 1866, where the values of A, B, C, D, Δ, E are all given; only those of A, B, C , are reproduced in the third edition, 1876.

It may be remarked, that for the general form we have $A = ag - 6bf + 15ce - 10d^2$, and that B is the determinant

$$B = \begin{vmatrix} a, & b, & c, & d \\ b, & c, & d, & e \\ c, & d, & e, & f \\ d, & e, & f, & g \end{vmatrix},$$

containing 24 terms. C and Δ are complicated forms, the latter of them containing 246 terms. But writing $c = 0, e = 0$, there is a great reduction; we have

$A =$	$B =$	$C =$	$D =$
$ag + 1$	$ad^2g - 1$	$a^2d^2f + 1$	$ag^5 + 1$
$bf - 6$	$bf^2 + 1$	$, df^2 + 4$	
$d^2 - 10$	$a^2b^2fg - 2$	$abd^2fg + 12$	$a^2b^2fg^4 - 30$
	$d^4 + 1$	$, d^4g - 20$	$, , d^2g^3 - 300$
		$a^2b^3df + 4$	$, , d^4g^2 - 2500$
		$, , bf^3 + 8$	$, , d^6g + 3125$
		$, , b^3d^3 - 24$	$, , df^3g^3 - 15$
		$, , bd^3f + 24$	$, , f^4 - 4800$
		$, , d^5 - 8$	$, , bdf^3g^2 + 7500$
			$, , bdf^2g \dots$
			$, , bd^3f^2 \dots$
			$\dots$
			$, , d^6g + 30000$
			$, , d^8g + 50000$
			$a^2b^2dg^4 - 2500$
			$, , b^2d^2fg^3 \dots$
			$, , b^2d^3g^2 - 171300$
			$, , b^2d^4fg - 240000$
			$, , bd^5g^2 + 780000$
			$, , b^2d^5 + 1200000$
			$, , d^7 - 1000000$
			$, , d^8 - 1600000$
			$, , d^9 - 7500$
			$\dots - 11520$
			ab^3dfg^3
			$+ 50000$
			$, , b^3f^4g$
			$, , b^3d^4fg^2 + 83200$
			$a^2b^4g^4 - 3125$
			$, , b^4d^2fg^2 - 240000$
			$, , b^4f^4 - 331776$
			$, , b^4d^2f^3 + 1200000$
			$, , b^4d^4f^2 + 1843200$
			$, , b^4d^6fg - 1600000$
			$, , b^4d^8 - 2560000$

It is clear that these are all functions of ag, bf, d^2 and $a^2f^2 + b^2g^2$, say of α, β, δ and ϕ . In fact, A and B are functions of α, β, δ ; C contains two terms, coefficient 4, which are $\pm\sqrt{\delta} \cdot 4 \cdot \phi$; Δ contains two terms

$$-3125(a^4f^4 + b^4g^4),$$

which are $= -3125(\phi^2 - 2\alpha^2\beta^2)$; and also several pairs of terms, each which pair contains the factor ϕ . We thus have

$A =$	$B =$	$C =$	$\Delta =$
$\alpha + 1$	$\alpha\delta - 1$	$\alpha^2\delta + 1$	$\alpha^4 + 1$
$\beta - 6$	$\beta^2 + 1$	$\alpha\beta\delta + 12$	$\alpha^3\beta - 30$
$\delta - 10$	$\beta\delta - 2$	$\alpha\delta^2 - 20$	$\alpha^3\delta - 300$
	$\delta^2 + 1$	$\beta^3 + 8$	$\alpha^2\delta^2 + 5840 (= 6250 - 410)$
		$\beta^2\delta - 24$	$\alpha^2\beta^2 - 15$
		$\beta\delta^2 + 24$	$\alpha^2\beta\delta - 4800$
		$\delta^3 - 8$	$\alpha^2\beta^3 + 3000$
			$\alpha^2\beta^2\delta - 171300$
			$\alpha^2\beta\delta^2 + 780000$
			$\alpha^2\delta^3 - 1000000$
			$\alpha^2\beta^4 - 11520$
			$\alpha^2\beta^3\delta + 83200$
			$\beta^4 - 331776$
			$\beta^3\delta + 1843200$
			$\beta^2\delta^2 - 2560000$
			$\phi\sqrt{\delta} \cdot \alpha^3 - 2500$
			, , $\alpha^2\beta - 7500$
			, , $\alpha\delta + 50000$
			, , $\beta^2 - 240000$
			, , $\beta\delta + 1200000$
			, , $\delta^2 - 1600000$
			$\phi^2 - 3125$

We have *ante*, the relation $\alpha + 9\beta - 20\delta = 0$, and using this to eliminate α , we have A, B, C as functions of β, δ, ϕ (that is, of bf, d^2 and $a^2f^2 + b^2g^2$). Effecting the substitution, we find the values of A, B, C without difficulty. As regards the value of Δ , this is

$$= -3125\phi^2 + 2\phi K\sqrt{\delta} + \text{terms without } \phi,$$

where

$$\begin{aligned} 2K = & -2500(81\beta^2 - 360\beta\delta + 400\delta^2) \\ & + 7500(-9\beta^2 + 20\beta\delta) \\ & + 50000(-9\beta\delta + 20\delta^2) \\ & - 240000(\beta^2) \\ & + 1200000(\beta\delta) \\ & - 1600000(\delta^2), \end{aligned}$$

or, reducing and dividing by 2,

$$K = -3125(60\beta^2 - 240\beta\delta + 256\delta^2).$$

The calculation of the terms without ϕ is much more laborious, but they come out

$$= -3125(60\beta^2 - 240\beta\delta + 256\delta^2)^2 \delta.$$

Hence the value of Δ is

$$\begin{aligned} \Delta = & -3125 \{ \phi^2 \\ & + 2\phi(60\beta^2 - 240\beta\delta + 256\delta^2)\sqrt{\delta} \\ & + (60\beta^2 - 240\beta\delta + 256\delta^2)^2 \delta \}, \end{aligned}$$

say this is

$$\Delta = -3125 h^2,$$

where

$$h = \phi + (60\beta^2 - 240\beta\delta + 256\delta^2)\sqrt{\delta},$$

that is,

$$= a^2 f^2 + b^2 g^2 + (60b^2 d^2 f^2 - 240bd^4 f + 256d^6).$$

The values of A, B, C , and the foregoing value of h then are

$A =$	$B =$	$C =$	$h =$
$\beta - 15$	$\beta^2 + 1$	$\beta^3 + \beta\delta$	$\beta^2\delta + 60$
$\delta + 10$	$\beta\delta + 7$	$\beta^2\delta + 51$	$\beta\delta\sqrt{\delta} - 240$
	$\delta^2 - 19$	$\beta\delta^2 + 84$	$\delta^2\sqrt{\delta} + 256$
		$\delta^3 - 8$	$\phi + 1$
		$\phi\sqrt{\delta} + 4$	

We may, if we please, regard β, δ, ϕ as irrational invariants of the sextic, viz. A, B, C being rational and integral functions of β, δ, ϕ , we have conversely β, δ, ϕ irrational functions of A, B, C ; and then the equation for h , say

$$\frac{h^2 (= \Delta)}{25\phi\sqrt{\delta}} = \phi + \sqrt{\delta}(60\beta^2 - 240\beta\delta + 256\delta^2),$$

is the invariantive relation which characterises the Jacobian sextic.

The expression for Δ in terms of A, B, C, D is

$$\Delta = A^5 - 375A^3B - 625A^2C + 3125D,$$

and it was in the foregoing investigation proper to use Δ in place of D . But I annex the value of D for the case in question $b = 0, f = 0$; and also its value in terms of $\alpha, \beta, \delta, \phi$. These are

$D =$	$D =$
$ag^5 - 1$	$\phi^2 - 1$
$, d^2fg - 12$	$\alpha^2f^2 + 2$
$, d^5 + 5$	$\alpha g^5 - 12$
$a^2b^2d^2f - 90$	$\beta - 72$
$, b^2df^2 - 48$	
$, bdf^2f + 246$	$bc^2f^4 + 168$
$, bd^3f^2 + 480$	$, \alpha^2$
$, d^6f - 258$	$, \beta^4 + 552$
$, d^8 - 432$	$\delta^2\alpha^2 + 5$
$ab^3df^4 - 12$	$, \alpha\beta + 246$
$, b^3f^5 - 12$	$\delta\alpha\beta^2 + 240$
$, b^3d^3f^2g + 168$	$, \beta^3 - 976$
$, b^3d^5fg + 240$	$\delta^2\alpha^2 - 258$
$, bd^6fg - 168$	$\delta\alpha\beta - 168$
$, d^8g - 228$	$, \beta^2 + 336$
$a^2b^4f^6 - 1$	$\delta^3\alpha - 228$
$, b^4d^2f^4 - 48$	$\delta\beta - 408$
$, b^4d^4f^2 - 72$	$\delta^4 - 240$
$, b^4d^4f^2 + 480$	$\phi\sqrt{\delta}\alpha\beta - 12$
$, b^4d^6f + 552$	$, \beta^2 - 48$
$, b^4d^8 - 432$	$\phi\sqrt{\delta}\beta - 480$
$, b^4d^8f^2 - 976$	$\phi\sqrt{\delta} - 432$
$, b^4d^6f^4 + 336$	
$, b^6d^8 + 408$	
$, d^{10} - 248$	

The Group of the Jacobian Sextic.

The solution of the Jacobian sextic equation depends upon that of a quintic; in fact, calling the roots $z_1, z_2, z_3, z_4, z_5, z_6$, then there exists a quintic having the roots

$$\begin{aligned} &\sqrt{(z_1 - z_2.z_1 - z_3.z_4 - z_5)}, \\ &\sqrt{(z_2 - z_4.z_2 - z_5.z_3 - z_6)}, \\ &\sqrt{(z_3 - z_5.z_4 - z_6.z_1 - z_2)}, \\ &\sqrt{(z_4 - z_6.z_5 - z_1.z_2 - z_3)}, \\ &\sqrt{(z_5 - z_1.z_6 - z_2.z_3 - z_4)}, \end{aligned}$$

the coefficients of which are rational functions of the coefficients a, b, d, f, g , and of the fourth root of the discriminant, i.e. $\sqrt[4]{h}$. But the meaning of this has not, so far as I am aware, been noticed. Passing to the quintic whose roots are the squares of the foregoing values, i.e. $z_1 - z_2, z_1 - z_3, z_4 - z_5$, &c., the coefficients are here rational functions of a, b, d, f, g and h ; that is, they are rational functions of a, b, d, f, g . The symmetrical functions of these roots $z_1 - z_2, z_1 - z_3, z_4 - z_5$, &c., are thus rational functions of the coefficients of the sextic; each such rational function is a 12-valued function of $z_1, z_2, z_3, z_4, z_5, z_6$, invariable by all the substitutions of a group of 60 substitutions; and therefore also every like 12-valued function of the roots $z_1, z_2, z_3, z_4, z_5, z_6$ is invariable by the substitutions of this group of 60; or, in other words, this group of 60 is the group of the Jacobian sextic equation.

I write for convenience, in this section only,

$$z_1, z_2, z_3, z_4, z_5, z_6 = f, a, b, c, d, e;$$

and writing further ab for shortness instead of $a - b$, &c., (so that of course $ba = -ab$), and putting

$$B = -ab.cd.ef, \quad C = -ac.bf.de, \quad D = ad.bc.ef, \quad E = ae.bd.cf, \quad F = af.be.cd,$$

then the five functions are B, C, D, E, F , and the group of 60 which leaves unaltered every symmetrical function of these functions is made up of the substitutions

1.			1	
$ab \cdot ce,$	$ab \cdot df,$	$ce \cdot df,$	15	
$ac \cdot bf,$	$ac \cdot de,$	$bf \cdot de,$		
$ad \cdot be,$	$ad \cdot ef,$	$be \cdot ef,$		
$ae \cdot bd,$	$ae \cdot cf,$	$bd \cdot cf,$		
$af \cdot bc,$	$af \cdot cd,$	$bc \cdot cd,$		
$abcde,$	$acebd,$	$adbec,$	$aedbc,$	24
$afbce,$				
$abdef,$	$abefc,$	$ae fcb,$	$aecbf,$	
$afced,$				
$afdbc,$	$adfbe,$	$aebfd,$	$afedb,$	
$bdcef,$				
	$acdf e,$	$ae fde,$	$adecf,$	
	$adcfb,$	$ab fed,$	$ac bdf,$	
	$bc fde,$	$bedfe,$	$bfecd,$	
$abc \cdot dfe,$	$acb \cdot def,$		20	
$abd \cdot cfe,$	$adb \cdot cef,$			
$abe \cdot cfd,$	$aeb \cdot cdf,$			
$abf \cdot ced,$	$afb \cdot cde,$			
$acd \cdot bef,$	$adc \cdot bfe,$			
$ace \cdot bfd,$	$aec \cdot bdf,$			
$acf \cdot bed,$	$afc \cdot bde,$			
$ade \cdot bfc,$	$aed \cdot bcf,$			
$adf \cdot bce,$	$afd \cdot bce,$			
$ae f \cdot bcd,$	$afe \cdot bcd,$		60	

where the symbols ab , $abcde$, abc , &c. denote cyclical substitutions. It is easy to verify that each of these substitutions does in fact merely permute B, C, D, E, F ; thus

$$\begin{array}{rccccc}
 & B & C & D & E & F \\
 abcde \text{ on } & -ab.ce.df, & -ac.bf.de, & ad.bc.ef, & ae.bd.cf, & af.be.cd \\
 = & -bc.da.ef, & bd.cf.ea, & be.cd.af, & ba.ce.df, & bf.ca.de \\
 = & +ad.bc.ef, & ae.bd.cf, & -af.be.cd, & -ab.ce.df, & ac.bf.de \\
 = & D & E & F & B & C,
 \end{array}$$

which (expressed as a cyclical substitution) is $= BDFCE$, and so in other cases.

We may to the foregoing 60 substitutions join the 60 other substitutions:

$$\begin{array}{lll}
 cdef, & cfed, & 30 \\
 bdf e, & bef d, & \\
 bcef, & bfce, & \\
 bcdf, & bf dc, & \\
 bced, & bdec, & \\
 aedf, & afde, & \\
 acef, & afec, & \\
 acfd, & adfc, & \\
 adce, & aecd, & \\
 & aefb, & \\
 & afbd, & \\
 abfe, & & \\
 adbf, & & \\
 abcd, & adcb, & \\
 abcf, & afcb, & \\
 aebe, & acbe, & \\
 abdc, & acdb, & \\
 ab \cdot cd \cdot ef, & ab \cdot ef \cdot de, & 10 \\
 ac \cdot bd \cdot ef, & ac \cdot be \cdot df, & \\
 ad \cdot be \cdot cf, & ad \cdot bf \cdot ce, & \\
 ae \cdot bc \cdot df, & ae \cdot bf \cdot cd, & \\
 af \cdot bc \cdot de, & af \cdot bd \cdot ce, & \\
 abcefd, & adfecb, & 20 \\
 abfdec, & acedfb, & \\
 abecdf, & af dceb, & \\
 abdfce, & aecfdb, & \\
 acfbde, & aedbf c, & \\
 acbfed, & ade fbc, & \\
 acdebf, & afbedc, & \\
 adbcfe, & aefcbd, & \\
 adebcf, & afebcd, & \\
 ae bdcf, & afcdbe, & 60
 \end{array}$$

each of which changes B, C, D, E, F into a permutation of $-B, -C, -D, -E, -F$.

The 60 and 60 substitutions form together a group of 120 substitutions, which leave unaltered any even symmetrical function of B, C, D, E, F , or say any symmetrical function of B^2, C^2, D^2, E^2, F^2 ; such a function is thus a 6-valued function of a, b, c, d, e, f , viz. it is Serret's 6-valued function of 6 letters.

Transformation of the Jacobian Sextic into the Resolvent Sextic of a special quintic equation.

Starting from the Jacobian Sextic Equation

$$(a, b, 0, d, 0, f, g \mid z, 1)^6 = 0,$$

$ag + 9bf - 20d^2 = 0$, I effect upon it the Tschirnhausen transformation

$$X = -az^2 - 6bz - 10d;$$

which, it may be remarked, is a particular case of the Tschirnhausen-Hermite form

$$\begin{aligned} X(= A + B) &= (az + b)B + (az^2 + 6bz + 5c)C + (az^3 + 6bz^2 + 15cz + 10d)D \\ &+ (az^4 + 6bz^3 + 15cz^2 + 20dz + 10e)E \\ &+ (az^5 + 6bz^4 + 15cz^3 + 20dz^2 + 15ez + 6f)F. \end{aligned}$$

Writing for convenience $Y = X + 10d$, $Z = X - 10d$, this is

$$+Y az^2 + 6bz - \dots = 0,$$

and we thence have

$$\begin{aligned} az^3 + 6bz^2 + \dots + Yz \cdot &= 0, \\ +az^4 + 6bz^3 \cdot + Yz^2 \cdot &= 0, \\ -Zz^2 \cdot -6fz - g &= 0, \\ -Zz^3 \cdot -6fz^2 - gz \cdot &= 0, \\ -Zz^4 \cdot -6fz^3 - gz^2 \cdot \cdot &= 0, \end{aligned}$$

or, eliminating, the resulting equation is

$$\begin{vmatrix} \cdot & a, & 6b, & \cdot & Y, & \cdot \\ \cdot & \cdot & a, & 6b, & \cdot & Y \\ a, & 6b, & \cdot & Y, & \cdot & \cdot \\ \cdot & \cdot & Z, & \cdot, & 6f, & g \\ \cdot & Z, & \cdot & 6f, & g, & \cdot \\ Z, & \cdot & 6f & g, & \cdot & \cdot \end{vmatrix} = 0.$$

The developed form is most easily obtained by expanding the determinant in the form

$$123 \cdot 456 - 456 \cdot 123; \text{ \&c.},$$

where the terms 123, &c., belong to the matrix

$$\begin{bmatrix} \cdot & a, & 6b, & \cdot & Y, & \cdot \\ \cdot & \cdot & a, & 6b, & \cdot & Y \\ a, & 6b, & \cdot & Y, & \cdot & \cdot \end{bmatrix}$$

and those of $\overline{123}$, &c., to the matrix

$$\begin{bmatrix} \cdot & \cdot & Z, & \cdot & 6f, & g \\ \cdot & Z, & \cdot & 6f, & g, & \cdot \\ Z, & \cdot & 6f, & g, & \cdot & \cdot \end{bmatrix}$$

The several terms are

$$\begin{array}{lll} 123 \cdot 456 & + - a^3 & \cdot - g^2 \\ & - 6a^2b & \\ -124 \cdot 356 & \cdot + & - 6fg^2 \\ +125 \cdot 346 & & \cdot \\ -126 \cdot 345 & - a^2Y & - g^2 - 216f^2 \\ & + - 36ab^2 & fZ - 216f \\ +134 \cdot 256 & - a^2Y & \cdot - fZ \\ & + - 6abY & \\ -135 \cdot 246 & + 6abY & - 6fgz \\ & - \cdot & - 6fgz \\ +136 \cdot 245 & & - 36f^2z \\ +145 \cdot 236 & & \\ -146 \cdot 235 & & \\ +156 \cdot 234 & & \\ -234 \cdot 156 & - a^2F - 216b^2 & - gZ^2 \\ +235 \cdot 146 & + 6abY & \\ -236 \cdot 145 & - - 36b^2Y & fZ \\ -245 \cdot 136 & - 36b^2Y & \\ +246 \cdot 135 & & - gZ \\ -256 \cdot 134 & & \\ +345 \cdot 126 & & 36fz \\ -346 \cdot 125 & & \\ +356 \cdot 124 & + aY^2 & gz^2 \\ -456 \cdot 123 & & \\ & - 6bY^2 & \cdot - 6fZ^2 \\ & Y^2 + -rt & - gZ^2 \\ & - 6bY^2 & - 6fZ^2 \\ & + & \\ & - - Y^3 & Z^3 \end{array}$$

Hence, collecting and reducing, the equation is

$$\begin{aligned} 0 &= Y^3Z^3. \\ &+ Y^2Z^2 \cdot (3ag + 72bf) \\ &+ YZ \cdot (3ag^2 + 36agbf + 1296b^2f^2) \\ &+ Y \cdot 216a^2f^3 \\ &+ Z \cdot -216b^3g^2 \\ &+ a^3g^3 - 36ag^2bf, \end{aligned}$$

where Y, Z denote $X + 10d, X - 10d$ respectively, and consequently $YZ = X^2 - 100d^2$. Hence, writing as before $\alpha, \beta, \delta, \phi$ to denote ag, bf, d^2 and $a^2f^2 + b^2g^2$ respectively, the result finally is

$$\left(1, \alpha + 3, \alpha^2 + 3, \alpha^3 + 216, \phi\sqrt{\delta} + 2160 \mid X, 1 \right)^6 = 0,$$

with the developed coefficients

$$\begin{aligned} &+0, \\ &\beta + 72, \\ &\alpha\beta + 36, \quad b^3f - 216, \\ &\delta - 300, \\ &\alpha\delta - 600, \\ &\beta^2 + 1296, \\ &\beta\delta - 14400, \\ &\delta^3 + 30000, \\ &-\alpha^3\beta - 36, \\ &\alpha^2\delta - 30, \\ &\alpha^2\delta - 360, \\ &\alpha\delta^2 + 30000, \\ &\beta^2\delta + 12960, \\ &\beta\delta^2 + 720000, \\ &\delta^3 - 1000000. \end{aligned}$$

where observe that the coefficient of the term in X is $216(a^2f^3 - b^3g^2) = 216\sqrt{\phi^2 - 4\alpha^2\beta^3}$.

We have as before $ag + 9bf - 20d^2 = 0$, that is, $\alpha + 9\beta - 20\delta = 0$; and using this equation to eliminate α , also in the constant term writing its value for ϕ in terms of h ,

$$\phi = h + (-60\beta^2 + 240\beta\delta - 256\delta^2)\sqrt{\delta},$$

the new equation is

$$\left(1, \beta - 9, \beta^2 - 243, -216\sqrt{h}, A\sqrt{\delta} + 432 \mid X, 1 \right)^6 = 0,$$

with developed coefficients

$$\begin{aligned} &\delta + 48, \quad \beta\delta - 1872, \\ &\beta^2 + 3840, \\ &\beta^3 - 729, \\ &\beta^2\delta + 4184, \\ &\beta\delta^2 - 11520, \\ &\delta^3 + 8292, \end{aligned}$$

where

$$\begin{aligned} \Lambda &= (h + (-60\beta^2 + 240\beta\delta - 256\delta^2)\sqrt{\delta})^2 - 4(-9\beta + 20\delta)^2\beta^3 \\ &= h^2 + 2h\sqrt{\delta} \begin{vmatrix} \beta^2 - 60 \\ \beta\delta + 240 \\ \delta^2 - 256 \end{vmatrix} - 4(\beta - 4\delta)^2(9\beta - 16\delta)^2. \end{aligned}$$

It is to be shown that this Tschirnhausen-transformation of the Jacobian sextic is, in fact, the resolvent sextic of the quintic equation

$$(a, 0, c, 0, e, f \mid x, 1)^5 = 0,$$

where

$$a = 1, \quad c = 2d, \quad e = -9bf + 36d^2, \quad f^2 = 216h.$$

I consider the general quintic $f^5 = (a, b, c, d, e, f \mid x, 1)^5 = 0$; taking the roots to be x_1, x_2, x_3, x_4, x_5 , and writing

$$\begin{aligned}\phi_1 &= 12345 - 24135, \\ \phi_2 &= 13425 - 32145, \\ \phi_3 &= 14235 - 43125, \\ \phi_4 &= 21435 - 13245, \\ \phi_5 &= 31245 - 14325, \\ \phi_6 &= 41325 - 12435,\end{aligned}$$

where 12345 is used to denote the function

$$= (x_1x_2 + x_2x_3 + x_3x_4 + x_4x_5 + x_5x_1)\sqrt{(20)},$$

(this numerical factor $\sqrt{(20)}$ being inserted for greater convenience), then the equation whose roots are $\phi_1, \phi_2, \phi_3, \phi_4, \phi_5, \phi_6$, which equation may be regarded as the resolvent sextic of the given quintic equation, is

$$\left(1, -5a^2x, 5a^4x^2, -\sqrt{(\square)} \cdot a^6, +5 \mid \phi, 1\right)^6 = 0,$$

with developed terms (selected):

$$\begin{aligned}&ae + 1 \\ &-a^2df + 1 \\ &+1 \\ &+a^3cf^2 + 1 \\ &a^2e^2 + 1 \\ &-2a^3def \\ &+ \&c. \\ &\&c.\end{aligned}$$

$D = +a^4f^4 \&c.$, the discriminant of the quintic: see p. 274* of my paper “On a new auxiliary equation in the theory of equations of the fifth order,” *Phil. Trans.*, t. CLI. (1861), pp. 263–276, [268].

I now write $b = 0$, $d = 0$, but, to avoid confusion again, write roman instead of italic letters, viz. I consider the resolvent sextic of the quintic equation

$$(a, 0, c, 0, e, f \mid x, 1)^5 = 0.$$

Many of the terms thus vanish, and the equation assumes the form

$$\left(1, -5a^4, 5a^8, -a^{12}\sqrt{\square}, +5 \mid \phi X, 1\right)^6 = 0,$$

with developed coefficients

$$\begin{aligned}&ae + 1, -3a^2cf^2 + 1 \\ &c^2 + 3, ac^2e - 2 \\ &c^2 + 15, ac^2f^2 + 1 \\ &a^3c^2e^2 - 11 \\ &ac^3e + 35 \\ &c^5 - 25\end{aligned}$$

and then if, as before,

$$\begin{aligned}a &= 1, c = 2d, e = -9bf + 36d^2, f^2 = 216h, \\ a &= 1, c = 2\sqrt{\delta}, e = -9\beta + 36\delta, f^2 = 216h,\end{aligned}$$

or say

*[This Collection, vol. iv., p. 321.]

this becomes identical with the foregoing Tschirnhausen-transformation equation; thus

$$\begin{aligned} -9ae + 3c^2 &= -9\beta + 36\delta + 12\delta, \\ &= \delta + 48; \end{aligned}$$

and similarly

$$\begin{aligned} 3a^3e^2 - 2ac^2e + 15c^4 &= \beta^2 + 243, \\ -\beta\delta - 1872, \\ +\delta^2 + 3840. \end{aligned}$$

So for the constant term, $+a^6cf^2$ gives the term $432h\sqrt{\delta}$, and $+a^6e^5$, &c., give the remaining terms $-729\beta^6$, &c. of the value in question.

It only remains to verify the equality of the coefficients of X ,

$$216\sqrt{h} = \sqrt{D} \quad \text{or} \quad 46656h = D.$$

Here D , the discriminant of the quintic $(a, 0, c, 0, e, f \mid x, 1)^5$, from the general form (see my Second Memoir on Quantics, [141], or Salmon's *Higher Algebra*, third edition, p. 209) putting therein $b = 0$, $d = 0$, is

$$\begin{aligned} D &= +a^4f^4 \cdot 1, \\ &+ a^3ce^2f + 160, \\ &a^3e^5 + 256, \\ &a^2c^3ef^2 - 1440, \\ &a^2c^3e^3 - 2560, \\ &ac^5f + 3456, \\ &ac^4e^2 + 6400, \end{aligned}$$

and writing for a, c, e, f their values $1, 2\sqrt{\delta}, 9(-\beta + 4\delta), 216h$, the value becomes

$$\begin{aligned} D &= (216)^2 \cdot h^2. \\ &+ 432h\sqrt{\delta} \cdot 12960 (\beta - 4\delta)^2 \\ &+ 34560 (\beta - 4\delta) \delta \\ &+ 55296 \delta^2 \\ &- 256 \cdot 9^5 \cdot -(\beta - 4\delta)^5 \\ &- 10240 \cdot 9^4 \cdot (\beta - 4\delta)^4 \delta \\ &- 102400 \cdot 9^3 \cdot (\beta - 4\delta)^3 \delta^2. \end{aligned}$$

The whole divides by $(216)^2$, and we thus obtain

$$\begin{aligned} \Lambda &= h^2 + 2h\sqrt{\delta} \cdot 60 (-\beta + 4\delta)^2 \cdot + (-\beta - 4\delta)^5 \cdot 324 (\beta - 4\delta)^4 \\ &\quad - 240 (-\beta + 4\delta) \delta \quad + 1440 (\beta - 4\delta) \delta \\ &\quad + 256 \delta^2 \quad + 1600 \delta^2, \end{aligned}$$

which is, in fact, equal to the foregoing value of Λ .

The conclusion is that, starting from the Jacobian sextic

$$f = (a, b, 0, d, 0, f, g \mid z, 1)^6 = 0,$$

where $ag + 9bf - 20d^2 = 0$, and effecting upon it the Tschirnhausen-transformation

$$X = -az^2 - 6bz - 10d,$$

so as to obtain from it a sextic equation in X , this sextic equation in X is the resolvent sextic of the quintic equation

$$(1, 0, c, 0, e, f \mid x, 1)^5 = 0,$$

where

$$c = 2d, \quad e = -9bf + 36d^2, \quad f = \sqrt{(216h)},$$

and, Δ being the discriminant of the Jacobian sextic, then

$$h = \frac{1}{5\sqrt{5}}\sqrt{(-\Delta)}, \quad = a^2f^2 + b^2g^2 + 60b^2d^2f^2 - 240bd^4f + 256d^6.$$

As to the subject of the present paper, see in particular Brioschi, “Ueber die Auflösung der Gleichungen vom fünften Grade,” *Math. Annalen*, t. XIII. (1878), pp. 109–160, and the third Appendix to his translation of my *Elliptic Functions*, Milan, 1880, each containing references to the earlier papers.